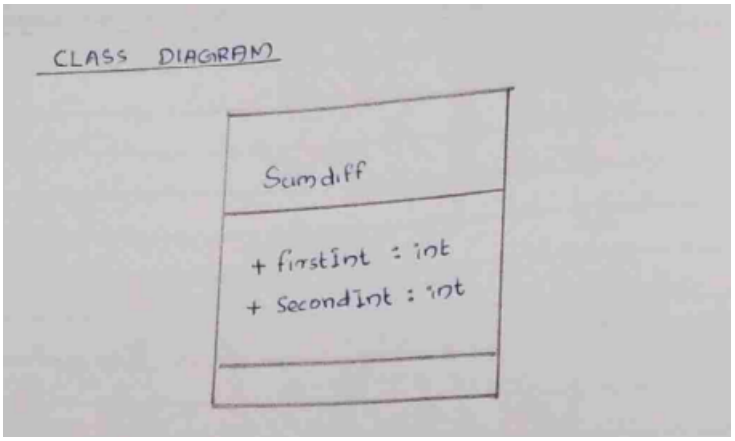
PART A - JAVA

1. **Write a program to find the sum, difference, product, quotient and remainder of two numbers passed as command line argument**.

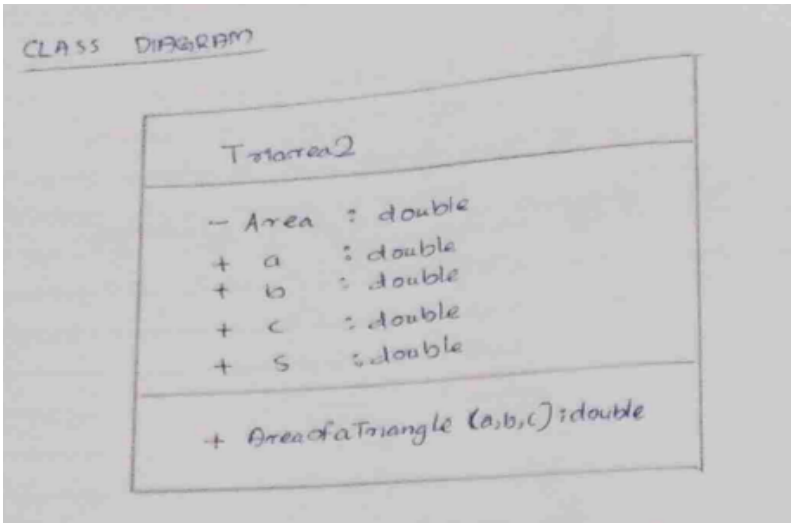



```
import java.util.Scanner;
    public class Sumdiff
    {

        public static void main(String[] args)
        {
            Scanner in = new Scanner(System.in);
            System.out.print("Input 1st integer: ");
            int firstInt = in.nextInt();
            System.out.print("Input 2nd integer: ");
            int secondInt = in.nextInt();

        System.out.printf("Sum of two integers: %d%n", firstInt + secondInt);
        System.out.printf("Difference of two integers: %d%n", firstInt - secondInt);
            System.out.printf("Product of two integers: %d%n", firstInt * secondInt);
            System.out.printf("Average of two integers: %.2f%n", (double) (firstInt +
    secondInt) / 2);
            System.out.printf("remainder of two integers: %d%n",firstInt %
    secondInt);
            System.out.printf("quotient of two integers: %d%n",firstInt / secondInt);
```

```
        System.out.printf("Distance of two integers: %d%n", Math.abs(firstInt - secondInt));

            System.out.printf("Max integer: %d%n", Math.max(firstInt, secondInt));

        System.out.printf("Min integer: %d%n", Math.min(firstInt, secondInt));

        }

}
```

```
C:\Program Files\Java\jdk1.8.0_171\bin>javac Sumdiff.java

C:\Program Files\Java\jdk1.8.0_171\bin>java Sumdiff
Input 1st integer: 12
Input 2nd integer: 14
Sum of two integers: 26
Difference of two integers: -2
Product of two integers: 168
Average of two integers: 13.00
remainder of two integers: 12
quotient of two integers: 0
Distance of two integers: 2
Max integer: 14
Min integer: 12

C:\Program Files\Java\jdk1.8.0_171\bin>
```

2. Given the sides of a triangle, write a program to check whether the triangle is equilateral, isosceles or scalene and find its area.



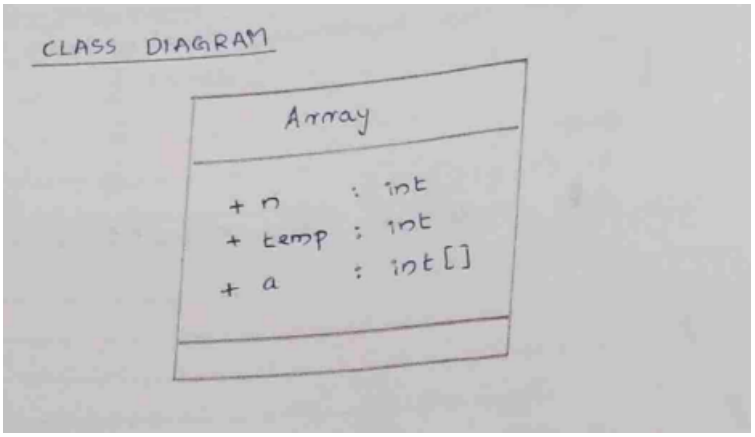

```
import java.util.Scanner;

public class  Triarea2 {

            private static Scanner sc;

   private static double Area;

            public static void main(String[] args) {

                        double a, b, c;
```

```
                sc = new Scanner(System.in);


                System.out.println("\n Please Enter Three sides of triangle: ");
                a = sc.nextDouble();
                b = sc.nextDouble();
                c = sc.nextDouble();
                  if(a==b && b==c)
        System.out.println("Equilateral");
        else if ((a==b && b!=c ) || (a!=b && c==a) || (c==b && c!=a))
        System.out.println("Isosceles");
      else if(a!=b && b!=c && c!=a)
        System.out.println("Scalene");
                Area = AreaofaTriangle(a, b, c);
                System.out.format("\n The Area of triangle = %.2f\n",Area);
          }
          public static double AreaofaTriangle( double a, double b, double c ) {
                double s;
                s = (a + b + c)/2;
                Area = Math.sqrt(s*(s-a)*(s-b)*(s-c));
                return Area;
          }
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Triarea2.java

C:\Program Files\Java\jdk-18.0.2\bin>java Triarea2

 Please Enter Three sides of triangle:
5
5
5
Equilateral

 The Area of triangle = 10.83

C:\Program Files\Java\jdk-18.0.2\bin>java Triarea2

 Please Enter Three sides of triangle:
5
6
7
Scalene

 The Area of triangle = 14.70

C:\Program Files\Java\jdk-18.0.2\bin>java Triarea2

 Please Enter Three sides of triangle:
5
6
5
Isosceles

 The Area of triangle = 12.00
```

3. Read an array of 10 or more numbers and write a program to find the

a) Smallest element in the array
b) Largest element in the array
c) Second largest element in the array



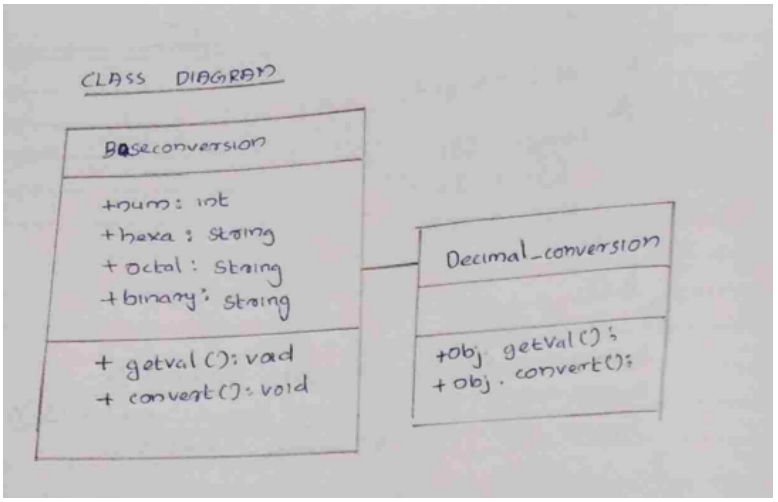

```
import java.util.Scanner;
public class Array
{
   public static void main(String[] args)
   {
      int n, temp;
      Scanner s = new Scanner(System.in);
      System.out.print("Enter no. of elements you want in array(Minimum 10):");
```

```
        n = s.nextInt();
        int a[] = new int[n];
        System.out.println("Enter all the elements:");
        for (int i = 0; i < n; i++)
        {
            a[i] = s.nextInt();
        }
        for (int i = 0; i < n; i++)
        {
            for (int j = i + 1; j < n; j++)
            {
                if (a[i] > a[j])
                {
                    temp = a[i];
                    a[i] = a[j];
                    a[j] = temp;
                }
            }
        }
        System.out.println("Largest:"+a[n-1]);
        System.out.println("Second Largest:"+a[n-2]);
        System.out.println("Smallest:"+a[0]);
    }
}
```

```
C:\Program Files\Java\jdk1.8.0_171\bin>javac Array.java

C:\Program Files\Java\jdk1.8.0_171\bin>java Array
Enter no. of elements you want in array(Minimum 10):10
Enter all the elements:
21
32
45
2
65
78
10
8
78
124
Largest:124
Second Largest:78
Smallest:2

C:\Program Files\Java\jdk1.8.0_171\bin>
```

4. Write a program to perform base conversion

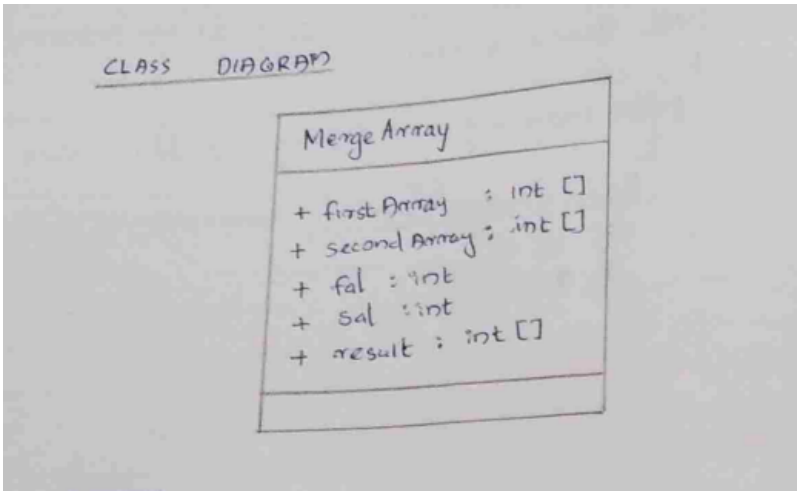
a) Integer to binary
b) Integer to Octal
c) Integer to Hexadecimal




```
import java.util.*;
class Baseconversion
{
Scanner scan;
int num;
void getVal()
{
System.out.println("Decimal to Hexadecimal,Octal and Binary");
scan= new Scanner(System.in);
System.out.println("\nEnter the number:");
num=Integer.parseInt(scan.nextLine());
}
void convert()
{
String hexa=Integer.toHexString(num);
System.out.println("HexaDecimal value is:" + hexa);
String octal=Integer.toOctalString(num);
System.out.println("Octal value is:" + octal);
String binary=Integer.toBinaryString(num);
System.out.println("Binary value is:"+binary);
}}

class Decimal_Conversion
{
public static void main(String args[])
{
Baseconversion obj=new Baseconversion();
obj.getVal();
obj.convert();
}
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Decimal_Conversion4.java

C:\Program Files\Java\jdk-18.0.2\bin>java Decimal_Conversion4
Decimal to Hexadecimal,Octal and Binary

Enter the number:
15
HexaDecimal value is:f
Octal value is:17
Binary value is:1111
```

5. Write a program to merge two arrays.



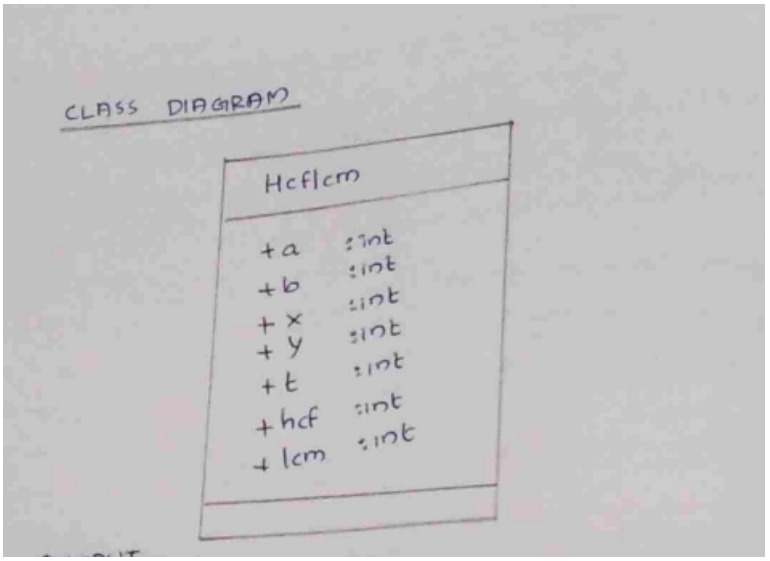

```
import java.util.Arrays;
import java.lang.*;
public class MergeArray1
{
public static void main(String[] args)
{
int[] firstArray = {23,45,12,78,4,90,1};
int[] secondArray = {77,11,45,88,32,56,3};
int fal = firstArray.length;
int sal = secondArray.length;
int[] result = new int[fal + sal];
System.arraycopy(firstArray, 0, result, 0, fal);
System.arraycopy(secondArray, 0, result, fal, sal);
```

```
System.out.println(Arrays.toString(result));
}
}
```

Output:

```
C:\Program Files\Java\jdk1.8.0_171\bin>javac MergeArray1.java

C:\Program Files\Java\jdk1.8.0_171\bin>java MergeArray1
[23, 45, 12, 78, 4, 90, 1, 77, 11, 45, 88, 32, 56, 3]
```

6. Java Programming Code to Find HCF LCM of Two Numbers

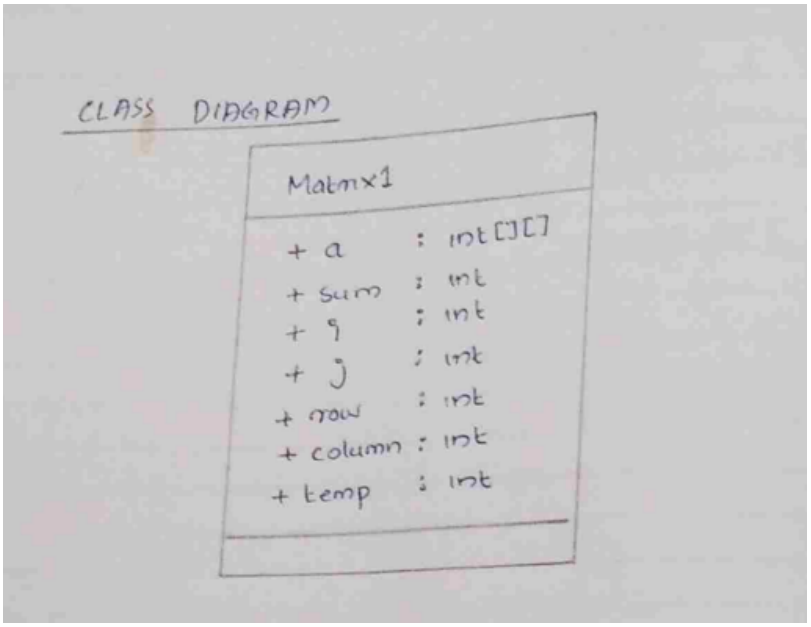



```
import java.util.Scanner;
public class Hcflcm
{
   public static void main(String args[])
   {
      int a, b, x, y, t, hcf, lcm;
      Scanner scan = new Scanner(System.in);

      System.out.print("Enter Two Number : ");
```

```
        x = scan.nextInt();

        y = scan.nextInt();

        a = x;

        b = y;

        while(b != 0)

        {

            t = b;

            b = a%b;

            a = t;

        }


        hcf = a;

        lcm = (x*y)/hcf;


        System.out.print("HCF = " +hcf);

        System.out.print("\nLCM = " +lcm);

    }

}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Hcflcm6.java

C:\Program Files\Java\jdk-18.0.2\bin>java Hcflcm6
Enter Two Number : 12
8
HCF = 4
LCM = 24
```

7. Write a program to find the trace and transpose of a matrix.



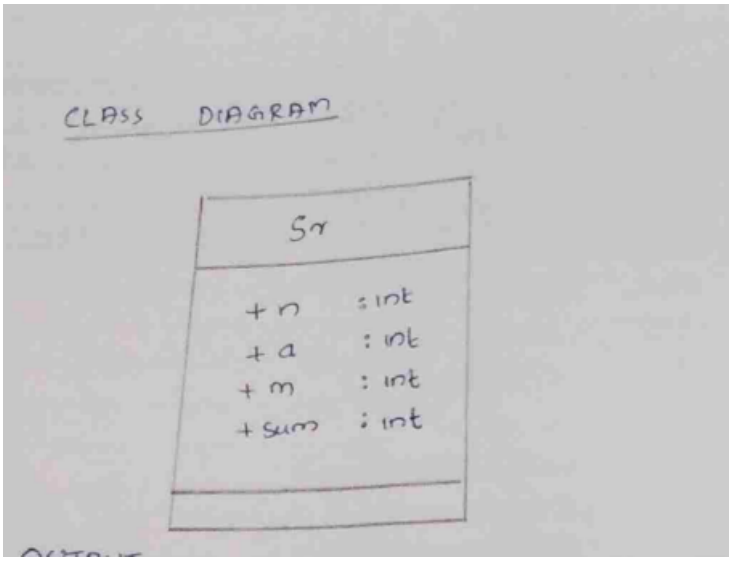

```
import java.util.*;
import java.io.*;
class Matrix1
{
public static void main(String args[])throws IOException
{
 DataInputStream in=new DataInputStream(System.in);
 int a[][]=new int[10][10];
 int sum=0;
 int i,j;
 int row,column,temp;
 System.out.println("Enter the no:f rows:");
 row=Integer.parseInt(in.readLine());
 System.out.println("Enter the no:of column:");
 column=Integer.parseInt(in.readLine());
 System.out.println("Enter the elements for the matrix:");
 for(i=0;i<row;i++)
 for(j=0;j<column;j++)
```

```
 a[i][j]=Integer.parseInt(in.readLine());
 System.out.println("The matrix is:");
 for(i=0;i<row;i++)
 {
 for(j=0;j<column;j++)
 System.out.print(a[i][j]+"\t");
 System.out.println();
 }
 System.out.println("The transpose of the matrix is:");
 temp=row;
 row=column;
 column=temp;
 for(i=0;i<row;i++)
 {
 for(j=0;j<column;j++)
 System.out.print(a[j][i]+"\t");
 System.out.println();
}
if(row!=column)
System.out.println("it is not a square matrix!!There is no trace for the matrix!!!");
else
{
for(i=0;i<row;i++)
for(j=0;j<column;j++)
if(i==j)
sum=sum+a[i][j];
System.out.println("The sum of the trace is:"+sum);
}
}
}
```

Output:

```
C:\Program Files\Java\jdk1.8.0_171\bin>javac Matrix1.java
Note: Matrix1.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

C:\Program Files\Java\jdk1.8.0_171\bin>java Matrix1
Enter the no:f rows:
3
Enter the no:of column:
3
Enter the elements for the matrix:
4
5
6
4
5
2
6
5
6
The matrix is:
4       5       6
4       5       2
6       5       6
The transpose of the matrix is:
4       4       6
5       5       5
6       2       6
The sum of the trace is:15

C:\Program Files\Java\jdk1.8.0_171\bin>
```

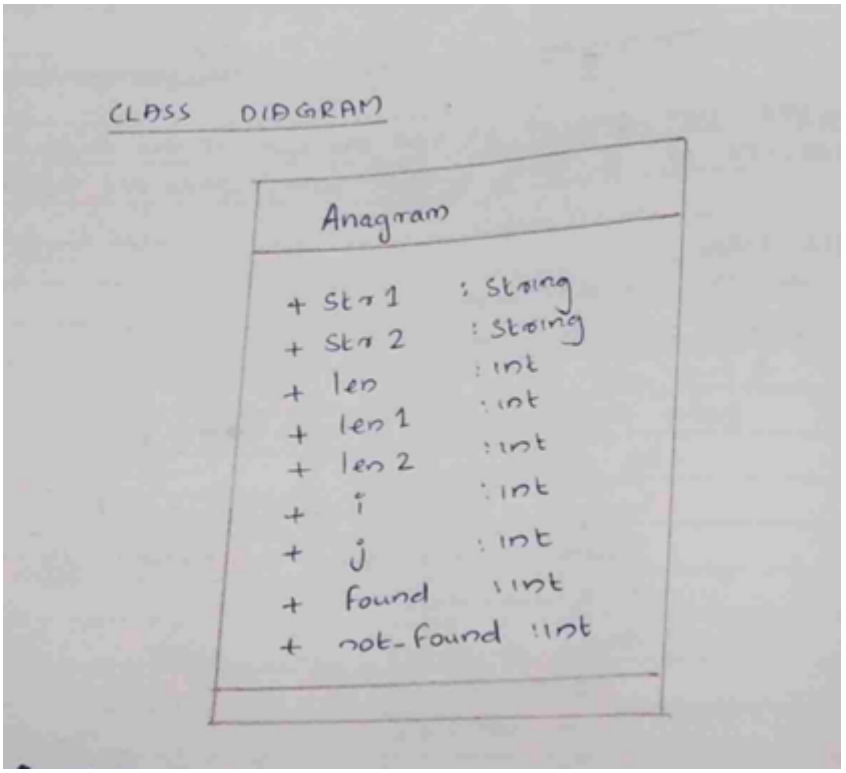
8. Write java program to find the sum of the digits and reverse of a given number using class and objects.



```
import java.util.Scanner;

public class Sr

{

   public static void main(String[] args)

   {

      int n, a, m = 0, sum = 0;

      Scanner s = new Scanner(System.in);

      System.out.print("Enter any number:");

      n = s.nextInt();

      do

      {

         a = n % 10;

         m = m * 10 + a;

         sum = sum + a;

         n = n / 10;

      }

      while( n > 0);

      System.out.println("Reverse:"+m);

      System.out.println("Sum of digits:"+sum);

   }

}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Sr8.java

C:\Program Files\Java\jdk-18.0.2\bin>java Sr8
Enter any number:369
Reverse:963
Sum of digits:18
```

9. Write a Java Programming Code to Check given string Anagram or Not. If the two strings are anagram to each other, then one string can be rearranged to form the other string. For Example: abc and cba are anagram.



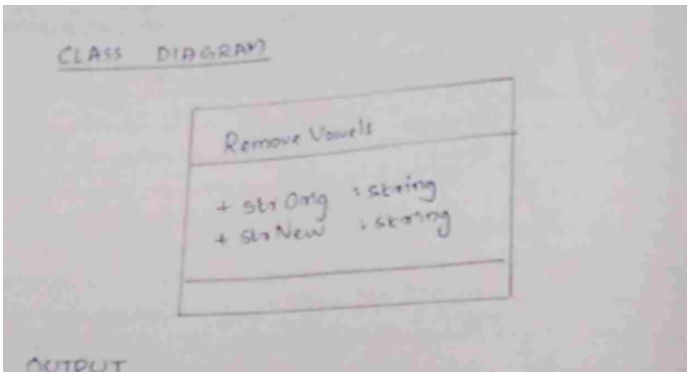

```
import java.util.Scanner;


public class Anagram

{

   public static void main(String[] input)

   {

      String str1, str2;

      int len, len1, len2, i, j, found=0, not_found=0;

      Scanner scan = new Scanner(System.in);


      System.out.print("Enter First String : ");

      str1 = scan.nextLine();

      System.out.print("Enter Second String : ");

      str2 = scan.nextLine();


      len1 = str1.length();

      len2 = str2.length();
```

```
 if(len1 == len2)
{
   len = len1;
   for(i=0; i<len; i++)
   {
      found = 0;
      for(j=0; j<len; j++)
      {
         if(str1.charAt(i) == str2.charAt(j))
         {
            found = 1;
            break;
         }
      }
      if(found == 0)
      {
         not_found = 1;
         break;
      }
   }
   if(not_found == 1)
   {
      System.out.print("Strings are not Anagram to Each Other..!!");
   }
   else
   {
      System.out.print("Strings are Anagram");
   }
}
else
{
```

```
        System.out.print("Both Strings Must have the same number of Character to be an Anagram");
    }
  }
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Anagram9.java

C:\Program Files\Java\jdk-18.0.2\bin>java Anagram9
Enter First String : earth
Enter Second String : heart
Strings are Anagram
C:\Program Files\Java\jdk-18.0.2\bin>java Anagram9
Enter First String : tool
Enter Second String : loop
Strings are not Anagram to Each Other..!!
C:\Program Files\Java\jdk-18.0.2\bin>java Anagram9
Enter First String : south
Enter Second String : out
Both Strings Must have the same number of Character to be an Anagram
```

10. Write a Java Program to remove all vowels from a string

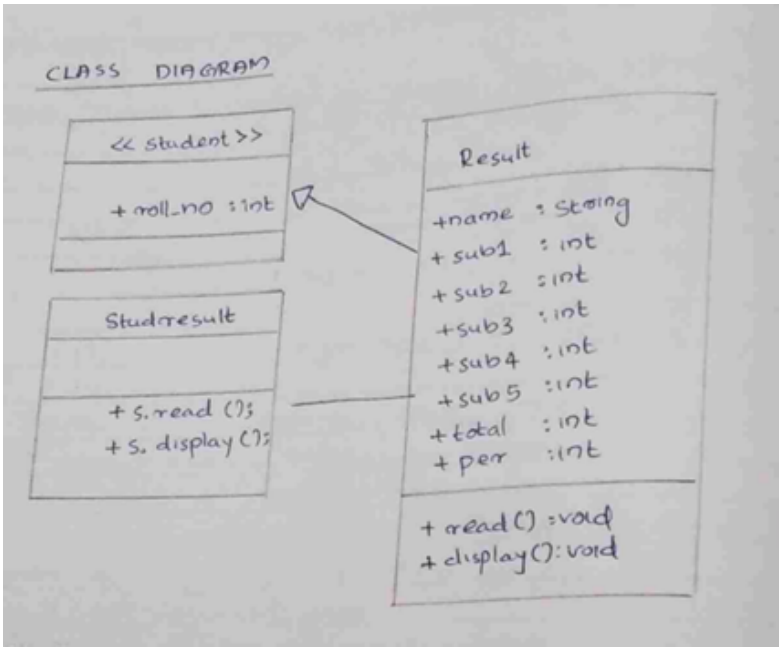



```
import java.util.Scanner;

public class Removevowels

{

  public static void main(String args[])

  {

    String strOrig, strNew;

    Scanner scan = new Scanner(System.in);


    System.out.print("Enter a String : ");
```

```
        strOrig = scan.nextLine();


        System.out.print("Removing Vowels from The String [" +strOrig+ "]\n");

        strNew = strOrig.replaceAll("[aeiouAEIOU]", "");


        System.out.print("All Vowels Removed Successfully..!!\nNow the String is :\n");


        System.out.print(strNew);

    }

}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Removevowels10.java

C:\Program Files\Java\jdk-18.0.2\bin>java Removevowels10
Enter a String : java programming
Removing Vowels from The String [java programming]
All Vowels Removed Successfully..!!
Now the String is :
jv prgrmmng
```

11. Create a class Student to read and display the student details. Create another class mark inherit from student to read marks of 5 subjects and find total and average. Write a Java program to display the result of a student.




```
import java.lang.*;

import java.io.*;
```

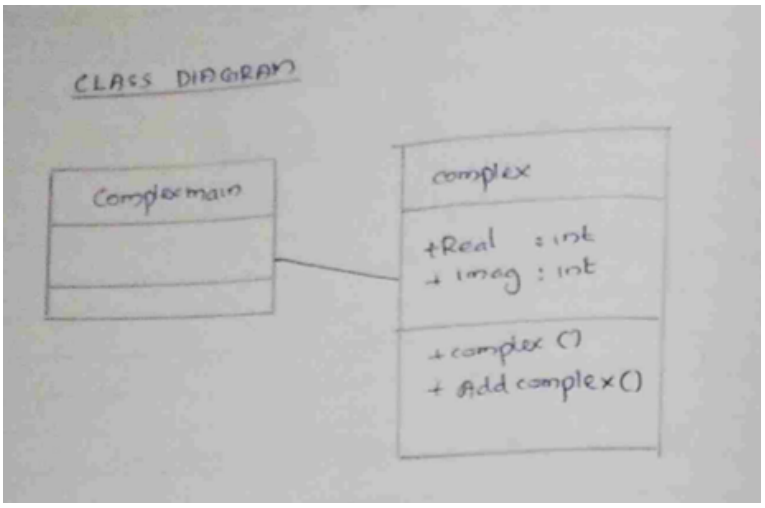
```
class Student
{
	int roll_no;
}
class Result extends Student
{
	String name;
	int sub1,sub2,sub3,sub4,sub5;
	int total;
	float per;
	void read() throws IOException
	{
		BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
System.out.println ("Enter Name of Student");
		name = br.readLine();
		System.out.println ("Enter Roll No. of Student");
		roll_no = Integer.parseInt(br.readLine());
		System.out.println ("Enter marks out of 100 of 1st subject");
		sub1 = Integer.parseInt(br.readLine());
		System.out.println ("Enter marks out of 100 of 2nd subject");
		sub2 = Integer.parseInt(br.readLine());
		System.out.println ("Enter marks out of 100 of 3rd subject");
		sub3 = Integer.parseInt(br.readLine());
		System.out.println ("Enter marks out of 100 of 4th subject");
		sub4 = Integer.parseInt(br.readLine());
		System.out.println ("Enter marks out of 100 of 5th subject");
		sub5 = Integer.parseInt(br.readLine());
	}
	void display()
	{
		total=sub1+sub2+sub3+sub4+sub5;
		per=(total*100)/500;
		System.out.println ("Roll No. = "+roll_no);
```

```
                System.out.println ("Name = "+name);
                System.out.println ("Marks of 1st Subject = "+sub1);
                System.out.println ("Marks of 2nd Subject = "+sub2);
                System.out.println ("Marks of 3rd Subject = "+sub3);
                System.out.println ("Marks of 4th Subject = "+sub4);
                System.out.println ("Marks of 5th Subject = "+sub5);
                System.out.println ("Total Marks = "+total);
                System.out.println ("Percentage = "+per+"%");
        }
}
class Studresult11
{
        public static void main(String args[]) throws IOException
        {
                Result s=new Result();
                s.read();
                s.display();
        }
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Studresult11.java

C:\Program Files\Java\jdk-18.0.2\bin>java Studresult11
Enter Name of Student
james
Enter Roll No. of Student
02
Enter marks out of 100 of 1st subject
85
Enter marks out of 100 of 2nd subject
78
Enter marks out of 100 of 3rd subject
94
Enter marks out of 100 of 4th subject
86
Enter marks out of 100 of 5th subject
91
Roll No. = 2
Name = james
Marks of 1st Subject = 85
Marks of 2nd Subject = 78
Marks of 3rd Subject = 94
Marks of 4th Subject = 86
Marks of 5th Subject = 91
Total Marks = 434
Percentage = 86.0%
```

12. Using class and objects, write a java program to find the sum of two complex numbers

(Hint: Use object as parameter to function).

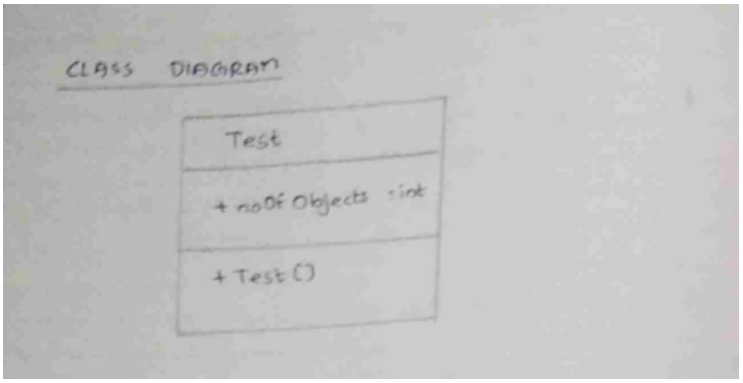



```
class Complex
  {
    int Real,Imag;
```

```
        Complex()
        {}

    Complex(int Real1,int Imag1)
    {
        Real=Real1;
        Imag=Imag1;
    }

    Complex AddComplex(Complex C1,Complex C2)
    {
        Complex CSum=new Complex();
        CSum.Real=C1.Real+C2.Real;
        CSum.Imag=C1.Imag+C2.Imag;
        return CSum;
    }
}

class Complexmain
{
    public static void main(String[] a)
    {
        Complex C1=new Complex(4,8);
        Complex C2=new Complex(5,7);
        Complex C3=new Complex();
        C3=C3.AddComplex(C1,C2);
        System.out.println("SUM:" + C3.Real +"+i" + C3.Imag);
    }
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Complexmain12.java

C:\Program Files\Java\jdk-18.0.2\bin>java Complexmain12
SUM:9+i15
```

13. Write a program to count and display total number of objects created to a class (Hint: static members).

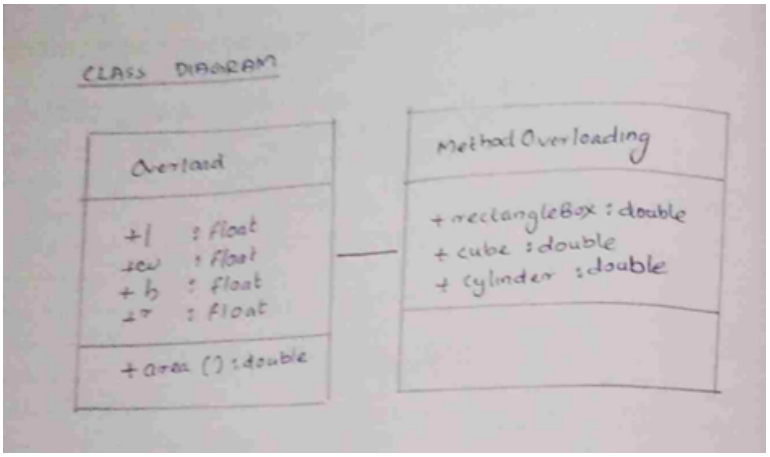



```
class Test

{

     static int noOfObjects = 0;

   {

      noOfObjects += 1;

   }

   public Test()

   {

   }

   public Test(int n)

   {

   }

   public Test(String s)

   {

   }

     public static void main(String args[])

   {
```

```
        Test t1 = new Test();

        Test t2 = new Test(5);

        Test t3 = new Test("GFG");

        System.out.println(Test.noOfObjects);

    }

}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac Test.java

C:\Program Files\Java\jdk-18.0.2\bin>java Test
3
```

14. Write a java program to find the volume of cube, rectangular box, cylinder using function overloading.

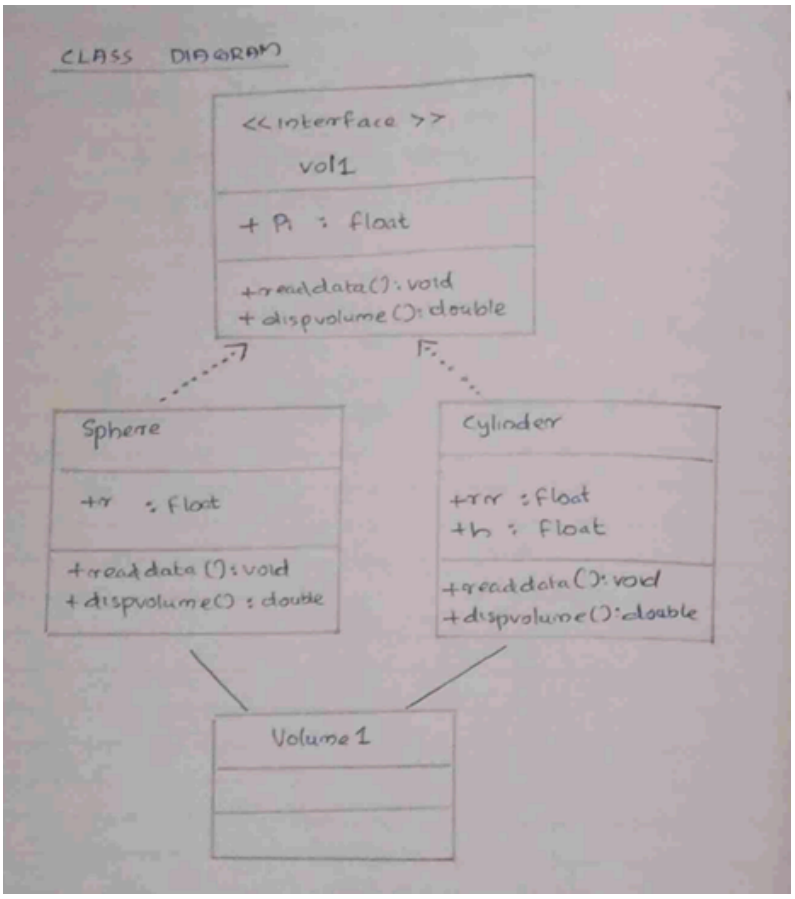



```
class Overload {
   double area(float l, float w, float h) {
      return l * w * h;
   }

   double area(float l) {
      return l * l * l;
   }

   double area(float r, float h) {
      return 3.1416 * r * r * h;
   }
}
public class MethodOverloading {
   public static void main(String args[]) {
      Overload overload = new Overload();
      double rectangleBox = overload.area(5, 8, 9);
      System.out.println("Area of ractangular box is " + rectangleBox);
      System.out.println("");
      double cube = overload.area(5);
```

```
        System.out.println("Area of cube is " + cube);

        System.out.println("");

        double cylinder = overload.area(6, 12);

        System.out.println("Area of cylinder is " + cylinder);

    }

}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac MethodOverloading.java

C:\Program Files\Java\jdk-18.0.2\bin>java MethodOverloading
Area of ractangular box is 360.0

Area of cube is 125.0

Area of cylinder is 1357.1712
```

15. Create an interface volume with member variable pi and methods readdata () and dispvolume (). Create two classes sphere and cylinder to implement this interface. Write a Java program to find the volume.

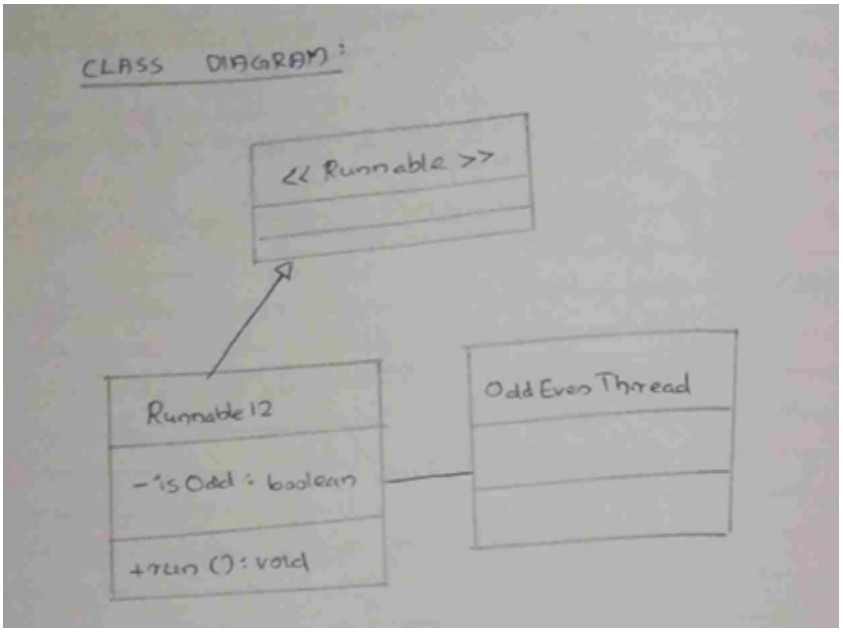


```
import java.*;

import java.util.Scanner;

interface vol1

{

final static float pi=3.14F;

void readdata();

double dispvolume();

}

class sphere implements vol1{

float r;

public void readdata()

{

Scanner scan=new Scanner(System.in);

System.out.println("enter the radius:");

r=scan.nextFloat();

}

public double dispvolume()

{

return(1.33*pi*r*r*r);

}

}

class cylinder implements vol1

{

float rr,h;

public void readdata()

{

Scanner scan1=new Scanner(System.in);

System.out.println("enter the radius:");

rr=scan1.nextFloat();

System.out.println("enter the height:");
```

```
h=scan1.nextFloat();
}
public double dispvolume()
{
return(pi*rr*rr*h);
}
}
class volume1
{
public static void main(String args[])
{
sphere s=new sphere();
vol1 vv;
vv=s;
System.out.println("the volume of sphere are:");
vv.readdata();
System.out.println("volume="+vv.dispvolume());
cylinder c=new cylinder();
vv=c;
System.out.println("the volume of cylinder are:");
vv.readdata();
System.out.println("volume="+vv.dispvolume());
}
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac volume1.java

C:\Program Files\Java\jdk-18.0.2\bin>java volume1
the volume of sphere are:
enter the radius:
3
volume=112.75740376710891
the volume of cylinder are:
enter the radius:
4
enter the height:
5
volume=251.20001220703125
```

16. Write a multi thread java program for displaying odd numbers and even numbers up to a limit (Hint: Implement thread using Runnable interface).

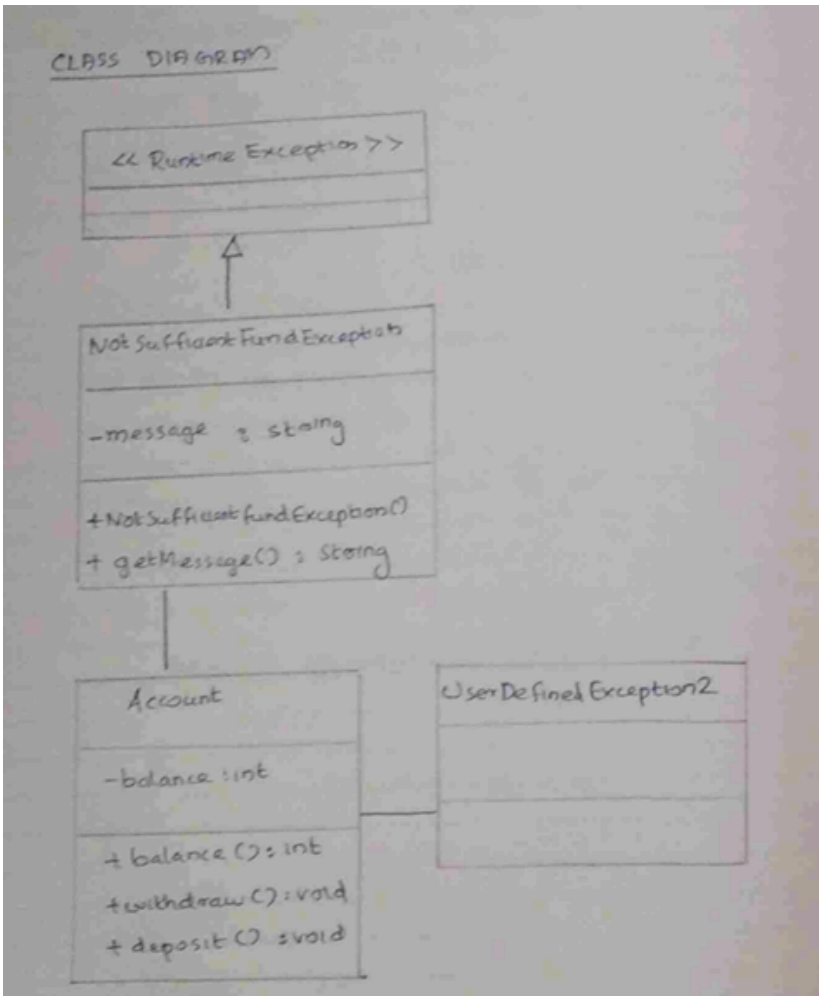



```
import java. lang.*;

public class OddEvenThread

 {

        public static void main(String[] args) {

                Runnable r1 = new Runnable12(true);

                Runnable r2 = new Runnable12(false);

                Thread t1 = new Thread(r1);

                Thread t2 = new Thread(r2);

                t1.start();
```

```
		t2.start();
	}
}

class Runnable12 implements Runnable {
	private boolean isOdd;

	Runnable12(boolean isOdd) {
		this.isOdd = isOdd;
	}

	public void run() {
		if (isOdd) {
			for (int i = 1; i <= 11; i += 2) {
				System.out.println(i);
			}
		} else {
			for (int i = 0; i < 11; i += 2) {
				System.out.println(i);
			}
		}
	}
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac OddEvenThread.java

C:\Program Files\Java\jdk-18.0.2\bin>java OddEvenThread
1
3
5
7
9
11
0
2
4
6
8
10
```

17. Create a class Account to deposit and withdraw money from a bank. Create a user defined exception "MinBalExp" to be invoked when the withdrawal amountis greater than balance.



import java.util.Scanner;

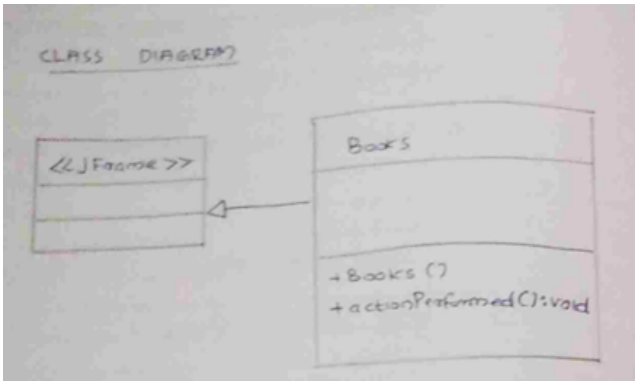
import java.lang.*;

```
class UserDefinedException2
 {
        public static void main(String args[])
        {
                Account acct = new Account();
                System.out.println("Current balance : " + acct.balance());
                System.out.println("Withdrawing 200");
                acct.withdraw(200);
                System.out.println("Current balance : " + acct.balance());
                acct.withdraw(1000);
   }
 }
  class Account
 {
 private int balance = 1000;
 public int balance()
 {
 return balance;
 }
 public void withdraw(int amount) throws NotSufficientFundException
 {
 if (amount > balance)
 {
 throw new NotSufficientFundException( String.format("Current balance %d is
less than requested amount %d", balance, amount));
 }
 balance = balance - amount;
 }
 public void deposit(int amount)
 {
```

```
if (amount <= 0)
{
throw new IllegalArgumentException( String.format("Invalid deposit amount
%s", amount));
}
}
}
class NotSufficientFundException extends RuntimeException
{
private String message;
public NotSufficientFundException(String message)
{
this.message = message;
}
public NotSufficientFundException(Throwable cause, String message)
{
super(cause);
this.message = message;
}
public String getMessage()
{
return message;
}
}
```

```
C:\Program Files\Java\jdk1.8.0_171\bin>javac UserDefinedException2.java

C:\Program Files\Java\jdk1.8.0_171\bin>java UserDefinedException2
Current balance : 1000
Withdrawing $200
Current balance : 800
Exception in thread "main" NotSufficientFundException: Current balance 800 is less than requested amount 1000
        at Account.withdraw(UserDefinedException2.java:26)
        at UserDefinedException2.main(UserDefinedException2.java:12)

C:\Program Files\Java\jdk1.8.0_171\bin>
```

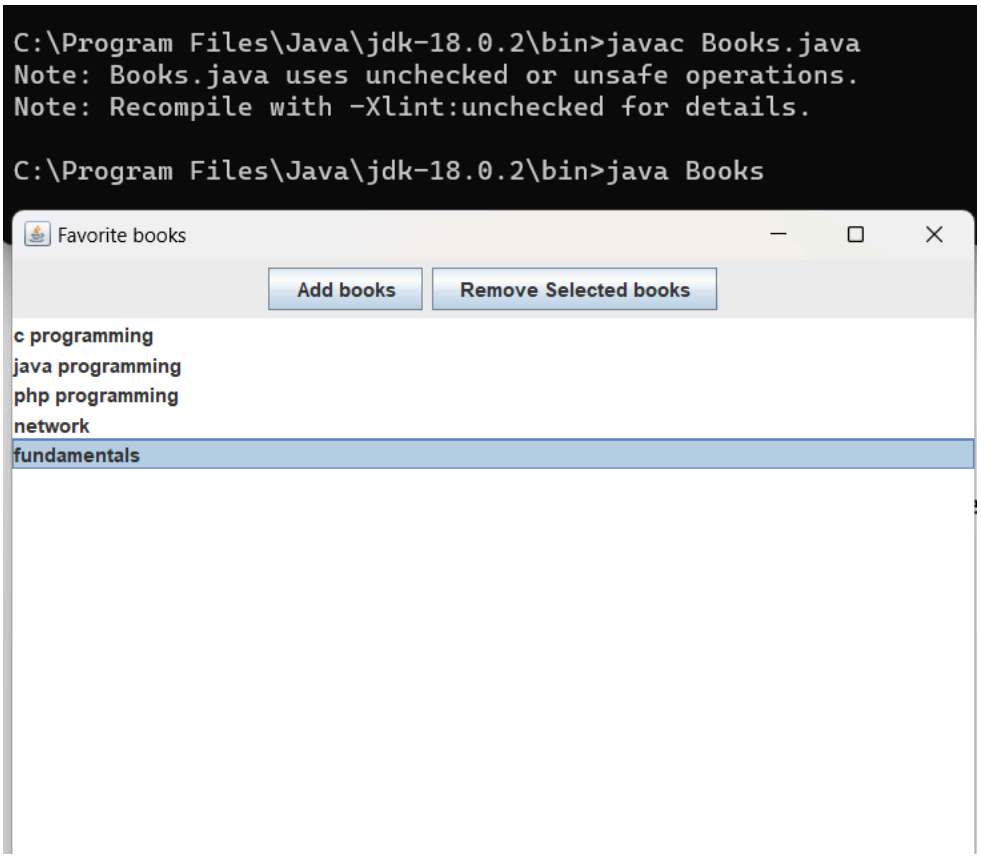
18. Create an AWT application to add, remove items in a list box.

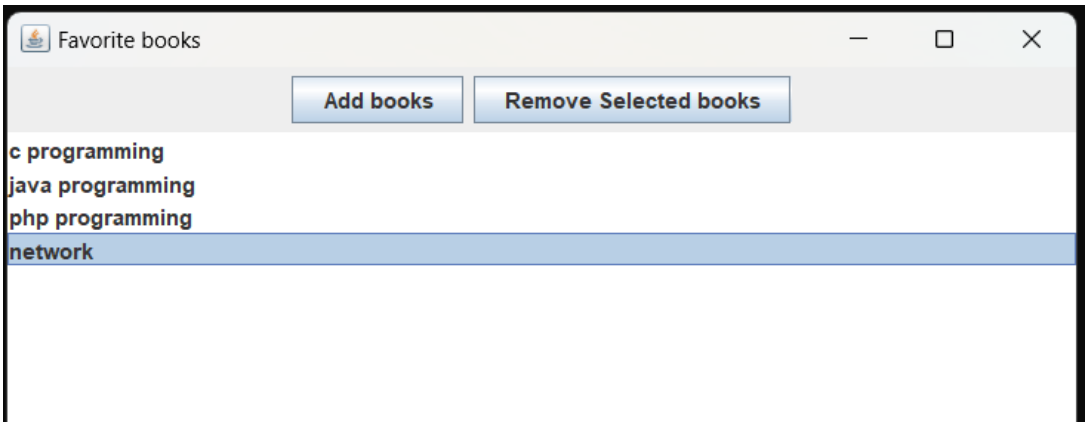


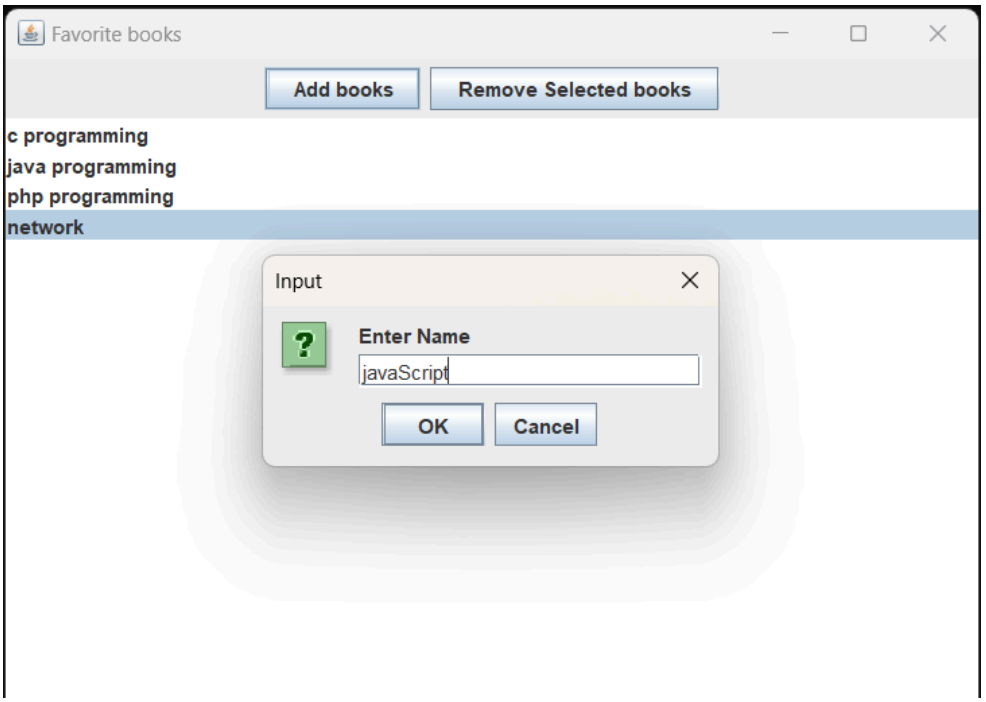

```
import java.awt.*;

import java.awt.event.*;

import javax.swing.*;

   public class Books extends JFrame {


    private DefaultListModel Books;

    private JList list;


    public Books()

      {

      super( "Favorite books" );

      Books = new DefaultListModel();

      Books.addElement( "c programming" );

      Books.addElement( "java programming" );

     Books.addElement( "php programming" );
```

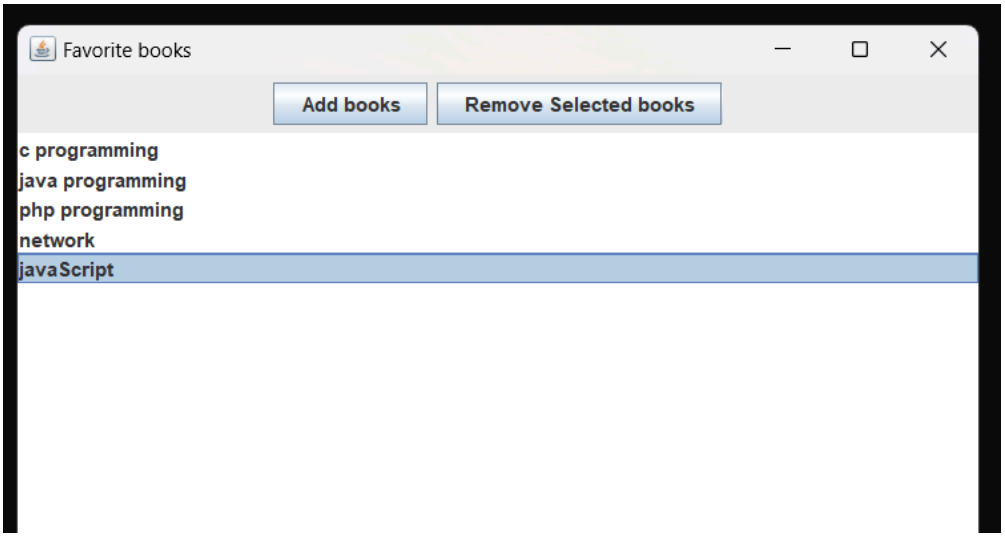
```
Books.addElement( "network" );
Books.addElement( "fundamentals" );
list = new JList( Books );
list.setSelectionMode(
ListSelectionModel.SINGLE_SELECTION );
JButton addButton = new JButton( "Add books" );
addButton.addActionListener(
    new ActionListener() {

    public void actionPerformed( ActionEvent event )
        {
        String name = JOptionPane.showInputDialog(
        Books.this, "Enter Name" );
        Books.addElement( name );
    }
}
);

JButton removeButton =
new JButton( "Remove Selected books" );
removeButton.addActionListener(
    new ActionListener() {

    public void actionPerformed( ActionEvent event )
        {
        Books.removeElement(
        list.getSelectedValue() );
    }
}
);
JPanel inputPanel = new JPanel();
```

```
        inputPanel.add( addButton );
        inputPanel.add( removeButton );
        Container container = getContentPane();
        container.add( list, BorderLayout.CENTER );
        container.add( inputPanel, BorderLayout.NORTH );
        setDefaultCloseOperation( EXIT_ON_CLOSE );
        setSize( 400, 300 );
        setVisible( true );

    }

        public static void main( String args[] )
        {
        new Books();
    }
}
```



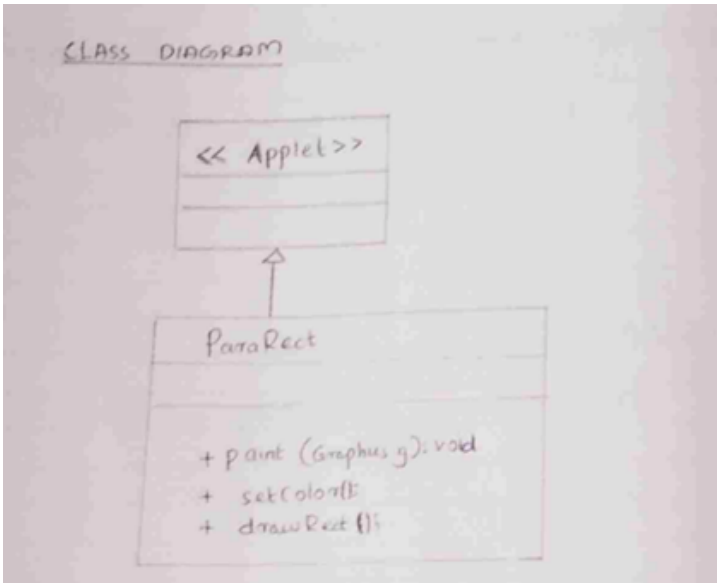

Favorite books
Add books
Remove Selected books
c programming
java programming
php programming
network


Favorite books
Add books
Remove Selected books
c programming
java programming
php programming
network
Input
Enter Name
javaScript
OK
Cancel


Favorite books
Add books
Remove Selected books
c programming
java programming
php programming
network
javaScript

---

19. Write an applet to display a rectangle with specified coordinate and colour passed as parameter from the HTML file.



ParaRect.html

```
<html>

<applet code="ParaRect.class" width=1200 height=1000>

<param name=x value=100>

<param name=y value=200>

<param name=color value= "FF0000">

</applet>

</html>
```

ParaRect.java

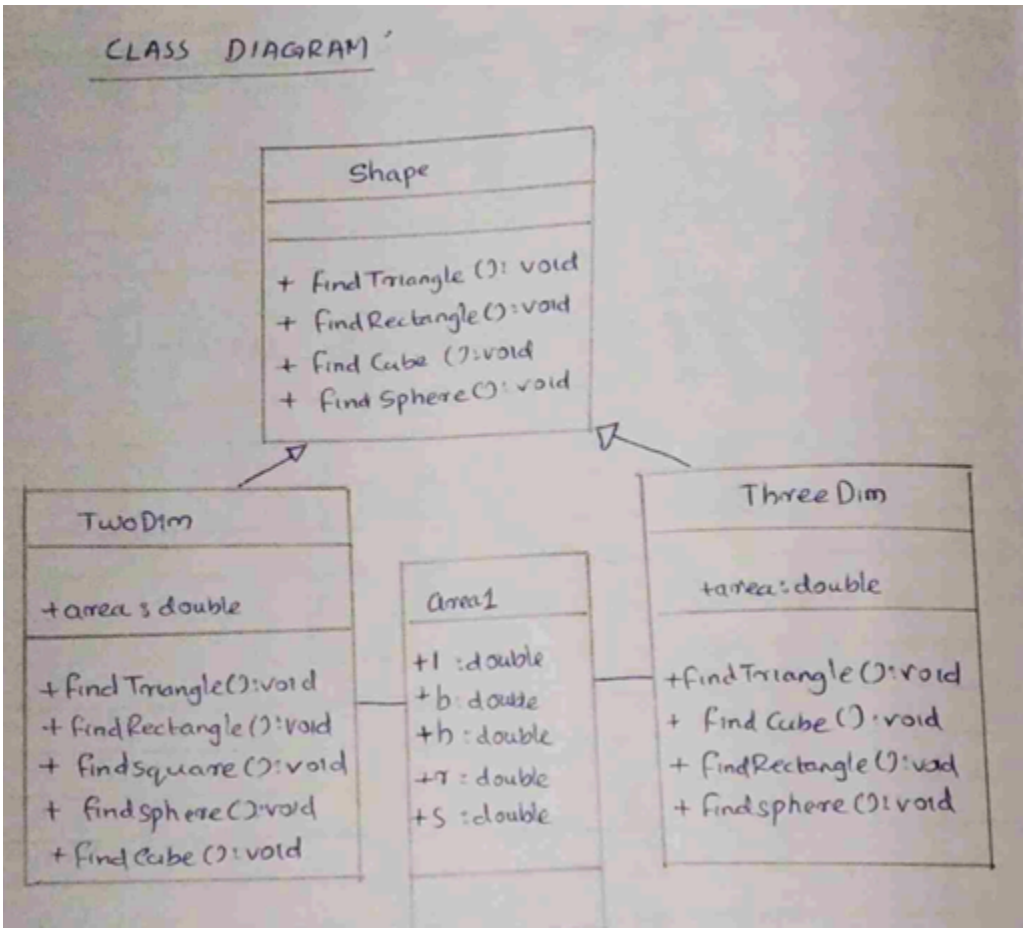
```
import java.awt.*;

import java.applet.*;

public class ParaRect extends Applet
```

```
{
  public void paint(Graphics g)
        {
    Color col =  new
    Color(Integer.parseInt(this.getParameter("color"),16));
    g.setColor(col);
          g.drawRect(Integer.parseInt(getParameter("x")),Integer.parseInt(getParameter("y")),150,100);
  }
}
```

```
C:\Program Files\Java\jdk-18.0.2\bin>javac ParaRect.java
```

```
C:\Program Files\Java\jdk-18.0.2\bin>appletviewer ParaRect.html
```

20. Create an abstract class shape and create TwoDim and ThreeDim as sub classes. Create classes square and triangle derived from TwoDim and sphere and cube derived from ThreeDim. Write a program to determinethe area of various shapes.



```
import java.util.Scanner;
abstract class shape
{
abstract void findTriangle(double b,double h);
abstract void findRectangle(double l,double b);
abstract void findCube(double s);
abstract void findSphere(double r);
}
class TwoDim extends shape
{
void findTriangle(double b,double h)
{
double area=(b*h)/2;
System.out.println("area of 2D Triangle:"+area);
}
void findRectangle(double l,double b)
{
double area=l*b;
System.out.println("Area of 2D Rectangle:"+area);
}
void findSquare(double s)
{
double area=s*s;
System.out.println("Area of 2D square:"+area);
}
void findSphere(double r)
{
double area=4*3.14*r*r;
System.out.println("Area of 2D sphere:"+area);
}
```

```
void findCube(double s)
{
double area=4*s*s;
System.out.println("Area of 2D cube:"+area);
}
}
class ThreeDim extends shape
{
void findTriangle(double b,double h)
{
double area=12*b*h;
System.out.println("area of 3D Triangle:"+area);
}
void findCube(double s)
{
double area=s*s*s;
System.out.println("Area of 3D cube:"+area);
}
void findRectangle(double l,double b)
{
double area=l*b;
System.out.println("Area of 3D Rectangle:"+area);
}
void findSphere(double r)
{
double area=(4/3)*3.14*r*r*r;
System.out.println("Area of 3D sphere:"+area);
}
}
class area1
```

```
{
public static void main(String args[])
{
double l,b,h,r,s;
TwoDim area=new TwoDim();
ThreeDim area1=new ThreeDim();
Scanner get=new Scanner(System.in);
System.out.print("\nEnter Base&Vertical Height of Triangle:");
b=get.nextDouble();
h=get.nextDouble();
area.findTriangle(b,h);
area1.findTriangle(b,h);
System.out.print("\nEnter Length&Breath of Rectangle:");
l=get.nextDouble();
b=get.nextDouble();
area.findRectangle(l,b);
System.out.print("\nEnter Side of a cube:");
s=get.nextDouble();
area.findCube(s);
area1.findCube(s);
System.out.print("\nEnter Radius of Sphere:");
r=get.nextDouble();
area.findSphere(r);
area1.findSphere(r);
}
}
```

Output :

```
C:\Program Files\Java\jdk1.8.0_171\bin>javac area1.java

C:\Program Files\Java\jdk1.8.0_171\bin>java area1

Enter Base&Vertical Height of Triangle:2
3
area of 2D Triangle:3.0
area of 3D Triangle:72.0

Enter Length&Breath of Rectangle:2
2
Area of 2D Rectangle:4.0

Enter Side of a cube:3
Area of 2D cube:36.0
Area of 3D cube:27.0

Enter Radius of Sphere:2
Area of 2D sphere:50.24
Area of 3D sphere:25.12
```

21. Create a database table employee (id, name, design, and dept.) and insert some records. Write a Java program to list the employee details using JDBC.

PART B- Java Script & PHP

Java Script

1. **Write a JavaScript program to check whether a given number is perfect, abundant or deficient. Use alert box to display the output.**

```
<html>
<h2>perfect abendent deficient<h2><br>
<body>
<form name=form1>
<input type="text" name="value1" size=10/>
<input type="button" value="calculate" onClick='check();'>
</form>
<script language="JavaScript">
function check()
{
var num=document.form1.value1.value;
```

```
var i,sum=0;
for(i=1;i<num;i++)
{
if(num%i==0)
sum=sum+i;
}
if(sum==num)
alert("perfect number");
else if(sum>num)
alert("abendent number");
else
alert("deficient number");
}
</script>
</body>
</html>
```

2. Design a form that accepts two integers. Provide 4 buttons for Add, Subtract, Multiply, Divide. Add JavaScript program to add, subtract, multiply and divide the given numbers when these buttons are clicked. Use output element to display the results.

```
<html>

<body>

<form name =form1>

enter the first number :

<input type = "text" name = "value1" size= 10/><br><br>

enter the second number:

<input type = "text" name = "value2" size= 10/><br><br>

<input type ="button" value="addition"onclick="add()">

<input type ="button" value="subtraction"onclick="sub()">

<input type ="button" value="multiplication"onclick="mul()">

<input type ="button" value="division"onclick="div()">

</form>

<script language="javaScript">

function add()

{

var num1=document.form1.value1.value;

var num2=document.form1.value2.value;
```

```
var sum=parseInt(num1)+parseInt(num2);
document.write("the sum is "+sum);
}
function sub()
{
var num1=document.form1.value1.value;
var num2=document.form1.value2.value;
var sub=num1-num2;
document.write("the sum is "+sub);
}
function mul()
{
var num1=document.form1.value1.value;
var num2=document.form1.value2.value;
var mul=num1*num2;
document.write("the sum is "+mul);
}
function div()
{
var num1=document.form1.value1.value;
var num2=document.form1.value2.value;
var div=num1/num2;
document.write("the sum is "+div);
}
</script>
</body>
</html>
```

3. Write a JavaScript program to store different colors in an array and change the background color of the page using this array elements

```
<html>

<body onmouseout="myArray(4)">

<script>

function myArray(n)

{

var backcolor=["#00ff00","#ff0000","#0000ff","#cce","#cd12ca"];

document.bgColor=backcolor[n];

}

</script>

<center>

<a href="#"onmouseenter="myArray(0)">green</a>

<a href="#"onmouseenter="myArray(1)">Red</a>

<a href="#"onmouseenter="myArray(2)">blue</a>

<a href="#"onmouseenter="myArray(3)">ash</a>

<h1>changing background color</h1>

</center>

</body>

</html>
```

4. Write a JavaScript program to create clock with a timing event.

```
<html>

<body onload="startTime()">

<h2>JavaScript Clock with event</h2>

<div id="txt"></div>

<script>

function startTime() {

  const today = new Date();
```

```
  let h = today.getHours();
  let m = today.getMinutes();
  let s = today.getSeconds();
  m = checkTime(m);
  s = checkTime(s);
  document.getElementById('txt').innerHTML =  h + ":" + m + ":" + s;
  setTimeout(startTime, 1000);
}
function checkTime(i) {
  if (i < 10) {i = "0" + i};  // add zero in front of numbers < 10
  return i;
}
</script>
</body>
</html>
```

5. Write a JavaScript Program to validate email based on the following condition

- **Mail id must contain the @ and . character**
- **There must be at least one character before and after the @.**
- **There must be at least two characters after . (dot).**

```
<html>
<head>
<title>email validation</title>
</head>
<script>
function validateemail()
{
var x=document.myform.email.value;
var atposition=x.indexOf("@");
var dotposition=x.lastIndexOf(".");
if (atposition<1 || dotposition<atposition+2 || dotposition+2>=x.length)
```

```
{
  alert("Please enter a valid e-mail address \n atpostion:"+atposition+"\n dotposition:"+dotposition);
  return false;
}
else alert(" a valid e-mail address ");
}
</script>
<body>
<form name="myform"  method="" action="" onsubmit="return validateemail();">
Email: <input type="text" name="email"><br/>

<input type="submit" value="register">
</form>
</body>
</html>
```

6. Write a JavaScript program for Password validation based on the following condition

- **Password and confirm password must be same**
- **Length of password must be greater than 8 characters**

```
<html>
<head>
<title> Verification of valid Password </title>
</head>
<script>
function verifyPassword() {
  var pw = document.getElementById("pswd").value;
  var pw2 = document.getElementById("pswd2").value;
   if(pw !=pw2&&pw!="")
    {
```

```
    document.getElementById("message").innerHTML = "passwordand confirmPassword are not same";

    return false;

            }

  else  if(pw.length < 8) {

    document.getElementById("message").innerHTML = "**Password length must be atleast 8
characters";

    return false;

  }

    else

            {

    alert("Password is correct");

  }

}

</script>

<body>

<center>

<h1 style="color:green">password validation</h1>

<form onsubmit ="return verifyPassword()">

<td> Enter user name </td>

<input type = "text" name= "user name">

<td> Enter Password </td>

<input type = "password" id = "pswd" value = "">

<td> Enter confirmPassword </td>

<input type = "password" id = "pswd2" value = "">

<span id = "message" style="color:red"> </span> <br><br>

<input type = "submit" value = "Submit">

<button type = "reset" value = "Reset" >Reset</button>

</form>

</center>

</body>

</html>
```

7. Write a JavaScript Program to create an Array and read values using Prompt popup box and display the sum of elements in an Alert Box

```
<html>

<head>

<title>create an array read values using prompt</title>

</head>

<body>

<script>

var arr=[];

var n=prompt('enter the number of element:');

for(var i=0;i<n;i++)

{

arr.push(prompt('enter the arraye element'+(i+1)));

}

var s=0;

for(i=0;i<n;i++)

{

s=s+parseInt(arr[i]);

}

alert('full array:'+arr.join(','));

alert('sum of array:'+s);

</script>

</body>

</html>
```

8. Change the text colour and back colour of a TextBox using onfocus and onBlur event

```
<html>

<head>

<title>onBlur onFocus</title>
```

```
</head>
<body>
<h2>change the text color and back color</h2>
<form name=form1>
<input type="text"id="txt"onFocus="txt_color()" onBlur="txt_back_color()"><br><br>
<input type="button" value="show">
</form>
<script language="Javascript">
function txt_color()
{
document.getElementById("txt").style.color="#FF0000";
document.getElementById("txt").style.background="#0000FF";
}
function txt_back_color()
{
document.getElementById("txt").style.color="#00FF00";
document.getElementById("txt").style.background="#FF0000";
}
</script>
</body>
</html>
```

9. Write a JavaScript program to display Capital of a country using onchange event. The country is selected from a select box and capital is displayed on a TextBox

```
<html>
<body>
<p>select a state from the list</p>
<select id="mySelect" onchange="display()">
<option value="Trivandrum">kerala</option>
<option value="Chennai">Tamil nadu</option>
<option value="Bengaluru">Karnadaka</option>
```

```
<option value="Panaji">Goa</option>

<option value="Patna">Bihar</option>

<option value="Chandigarh">Haryana</option>

<option value="Bhopal">Madhya pradesh</>

<option value="Imphal">Manipur</>

<option value="Shillong">Megalaya</>

</select>

<br>

<br>

<input type = "text" id="txt">

<script>

function display()

{

var x=document.getElementById("mySelect").value;

document.getElementById("txt").value=x;

}

</script>

</body>

</html>
```

10. Write a JavaScript program to find 1st January be a Sunday between a range of years.

```
<html>
<script type="text/javascript">
  function jansunday()
      {
  var y1 = Number(document.getElementById('year1').value);
  var y2 = Number(document.getElementById('year2').value);
  var res = " ";
  for (let year = y1; year <= y2; ++year)
      {
  const c_year=new Date(year, 0, 1);
  if (c_year.getDay()===0)
      {
  res +=year + " ," ;
```

```
    }
    }
        if (y1> y2)
                {
    document.getElementById('jan').innerHTML= "Year2 must be greater than
Year1";
    }
        else
        {
    if (res === " ")
        {
    document.getElementById('jan').innerHTML = `<p>No Year Exist</p>`;
    } else {
    document.getElementById('jan').innerHTML = `<p> Year having
       Sunday on 1st January are :${res}</p>`;
    }
    }
    }
</script>
<body>
    <label for="Year1">Year1:</label>
    <input type="number" id="year1" name="Year1"><br><br>
    <label for="Year2">Year2:</label>
    <input type="number" id="year2" name="Year2"><br><br>
    <button onClick="jansunday()">
       Get Result
    </button>
        <p id="jan"></p>
        </body>
</html>
```

11. Write a JavaScript program to check whether the given sides can form a triangle. If yes, find the type (isosceles, equilateral and scalene) and area of the triangle. Use prompt dialogue box to accept the sides.

PHP

1. Create a PHP program to display the bio data of a person by reading the personal details using an HTML page.

```
<html> <body>
<form action="" method="POST">

<H1><center>BIO-DATA</center></h1>
Name:<input type=text name="name1"><br>
Address: <textarea name=address1>
```

```
</textarea><br> Age: <input type=text
name=age1><br>
Mobile: <input type=text
name=mobile1><br> Email: <input
type=text name=email1><br>
Qualification:<input type=text name=qualification1><br>
<input type=submit value=Display ><br>
</form> </body> </html>

<?ph
p
if($_
POST
){
        echo
        "<h1><center>BIO-DATA</center></h1>";
        echo "Name: ".$_POST["name1"]."<br>";
        echo "Address: ".$_POST["address1"]."<br>";
        echo "Age: ".$_POST["age1"]."<br>";
        echo "Mobile
        ".$_POST["mobile1"]."<br>"; echo
        "Email ".$_POST["email1"]."<br>";
        echo "Qualification ".$_POST["qualification1"]."<br>";
}
?>
```

2. Write a PHP program to check whether a given number is perfect, abundant or deficient.

```
<html>
<body>
<h2>Enter the number</h2><br>
<form action="" method="post">
<input type="text"name="number"/>
<input type="submit"/>
</form>
</body>
</html>
<?php
if($_POST)
{
        $no=$_POST['number'];
        $sum=0;
        for($i=1;$i<$no;$i++)
        {
                if($no%$i==0)
                        $sum=$sum+$i;
        }
        if($sum==$no)
                echo"Perfect number";
        else if($sum>$no)
                echo"Abundant Number";
```

```
        else
                echo"Deficient Number";
}
?>
```

3. Create a login page using database.

```
<html>
<head>
</head>
<body>
<form action="" method="POST">
Username:<input type="text" name="usr"><br><br>
Password:<input type="password" name="pass"><br><br><br>
<input type="submit" value="login">
</form>
</body>
</html>
<?php
if($_POST)
{
$usr1=$_POST['usr'];
$paswd=$_POST['pass'];
$con=pg_connect("host=localhost dbname=loginform user=postgres
password=123");
if($con)
{
echo "Successfully Connected.....";
$qry="select username,password from logintable where username='$usr1' and
password='$paswd'";
$result=pg_query($con,$qry);
$nos=pg_num_rows($result);
if($nos)
echo "<br>$usr1,You Are Logged Successfully...";
else
echo "Login Denied";
}
}
?>
```

4. Create a table student with fields roll no, name, mark, grade. Insert some records in the table. Write a PHP program to display the mark list of a student by accepting the register no of the student.

```
<html>

<head>

<title>MARKLIST</title>

</head>

<body>

<h3><center>Marksheet</center></h3>
```

```
<form method="POST" action=" ">
<label>Regno:</label>
<input type="text"name="txtreg"/><br><br><br>
<input type="submit"value="show">
</form>
</body>
</html>
<?php
if($_POST)
{
$rg=$_POST['txtreg'];
echo $rg;
$con=pg_connect("host=localhost dbname=college user=postgres
password=123");
if($con)
echo"Successfully Connected...";
$qry="select * from stud where roll_no='$rg'";
$result=pg_query($con,$qry);
$nos=pg_num_rows($result);
while($row=pg_fetch_row($result))
{
echo"<br>\n";
echo"roll_no:$row[0] name:$row[1] mark:$row[2] grade:$row[3]";
}
}
?>
```

5. Write a PHP application to generate the pay slip of an employee by accepting name, basic salary and designation. The net salary will be calculated based on the following conditions.

Designation conveyance extra allowance
allowance
Manager 1000 500
Supervisor 750 200
Clerk 500 100
Peon 250
HRA – 25 %

Net = Gross – Income tax

Income tax
Gross upto 2500 0
Gross upto 4000 3%
Gross upto 5000 5%
Gross >5000 8%

Gross= basic + HRA + conveyance + extra

```
<html>
<head>
<title>Payroll page</title>
<center><h2>Payroll statments</h2></center>
</head>
<body>
<form action=""method="POST">
<table border="0">
<tr>
<td>Employee Name</td>
<td><input type="text"name="name"></td>
</tr>
<tr>
<td>Designation</td>
<td><select name="desig">
<option>(Please Select)</option>
<option>Manager</option>
<option>Supervisor</option>
<option>Clerk</option>
<option>Peon</option>
</select>
</td>
```

```
</tr>
<tr>
<td>Basic salary</td>
<td><input type="text"name="salary"></td>
</tr>
<tr>
<td>
</td>
<td><input type="submit"value="submit">
</td>
</tr>
</table>
</form>
</body>
</html>
<?php
if($_POST)
{
$ename=$_POST["name"];
$designation=$_POST["desig"];
$basic=$_POST["salary"];
if(is_numeric($basic))
{
if($basic!=NULL)
{
switch($designation)
{
case"Manager":
$hra=($basic*25)/100;
$conv=1000;
```

```
$extra=500;
$gross=$hra+$conv+$extra+$basic;
if($gross<=2000)
$intax=$gross;
elseif($gross>2000&&$gross<4000)
$intax=($gross*3)/100;
elseif($gross>4000&&$gross<5000)
$intax=($gross*5)/100;
elseif($gross>5000)
$intax=($gross*8)/100;
$net=$gross-$intax;
echo"<b>Employee Name:</b>$ename<br>";
echo"<b>Designation:</b>$designation<br>";
echo"<b>Salary:</b>$net<br>";
break;
case"Supervisor":
$hra=($basic*25)/100;
$conv=750;
$extra=200;
$gross=$hra+$conv+$extra+$basic;
if($gross<=2000)
$intax=$gross;
elseif($gross>2000&&$gross<4000)
$intax=($gross*3)/100;
elseif($gross>4000&&$gross<5000)
$intax=($gross*5)/100;
elseif($gross>5000)
$intax=($gross*8)/100;
$net=$gross-$intax;
echo"<b>Employee Name:</b>$ename<br>";
```

```
echo"<b>Designation:</b>$designation<br>";
echo"<b>Salary:</b>$net<br>";
break;
case"Clerk":
$hra=($basic*25)/100;
$conv=500;
$extra=100;
$gross=$hra+$conv+$extra+$basic;
if($gross<=2000)
$intax=$gross;
elseif($gross>2000&&$gross<4000)
$intax=($gross*3)/100;
elseif($gross>4000&&$gross<5000)
$intax=($gross*5)/100;
elseif($gross>5000)
$intax=($gross*8)/100;
$net=$gross-$intax;
echo"<b>Employee Name:</b>$ename<br>";
echo"<b>Designation:</b>$designation<br>";
echo"<b>Salary:</b>$net<br>";
break;
case"Peon":
$hra=($basic*25)/100;
$conv=250;
$extra=0;
$gross=$hra+$conv+$extra+$basic;
if($gross<=2000)
$intax=$gross;
elseif($gross>2000&&$gross<4000)
$intax=($gross*3)/100;
```

```
elseif($gross>4000&&$gross<5000)
$intax=($gross*5)/100;
elseif($gross>5000)
$intax=($gross*8)/100;
$net=$gross-$intax;
echo"<b>Employee Name:</b>$ename<br>";
echo"<b>Designation:</b>$designation<br>";
echo"<b>Salary:</b>$net<br>";
break;
default:
echo("Invalid Designation! tryagain.....");
}
}
}
}
?>
```

6. Write an HTML page to display a list of fruits in a list box. Write php program to display the selected fruits in a webpage.

```
<html>
<body>
<form action="" method="POST">
Choose your Favourite Fruit<br><br>
<select name="f">
<option value="....">(Please select)
<option value="Grape">Grape
<option value="banana">Banana
<option value="chicku">chicku
<option value="apple">Apple
<option value="orange">Orange
```

```
<option value="pineapple">Pineaplle
</select><br><br>
<input type="submit" value="select">
</form>
</body>
</html>
<?php
if($_POST)
{
        echo"<h2>You Have indicated that you like</h2>";echo $_POST["f"];
}
?>
```

7. PHP program to store current date-time in a cookie and display the Last visited date-time on the web page upon revisiting the same web page.

```
<html>
<body bgcolor="yellow">
<center><h2> Last visited time on the web page</h2></center>
<br>
<?php

$inTwoMonths = 60 * 60 * 24 * 60 + time();
setcookie('lastVisit', date("G:i - m/d/y"), $inTwoMonths);
if(isset($_COOKIE['lastVisit']))

{
$visit = $_COOKIE['lastVisit'];
echo "Your last visit was - ". $visit;
}
else
```

```
echo "You've got some stale cookies!";
?>
</body>
</html>
```

8. Create a table product with fields itemcode, itemname, unitprice. Write PHP program to insert 5 records into the table and display it in a table format.

```
<html>
<head>
<title>Product</title>
</head>
<body>
<h1><center>List of Products</center></h1>
<form method="POST" action="">
Item Code:<input type="text" name="icode"/><br><br><br>
Item Name:<input type="text" name="iname"/><br><br><br>
Unit Price:<input type="text" name="iprice"/><br><br><br>
<input type="submit" value="Display">
</form>
</body>
</html>
<?php
if($_POST)
{
$no=$_POST['icode'];
$name=$_POST['iname'];
$price=$_POST['iprice'];
$con=pg_connect("host=localhost dbname=product user=postgres
password=123");
if($con){
echo"Successfully Connected.....";
```

```
$qry="insert into
product(itemcode,itemname,unitprice)values($no,'$name',$price)";

$result=pg_query($con,$qry);

$qry1="select * from product";

$result1=pg_query($con,$qry1);

while($row=pg_fetch_row($result1)){

echo"<br>\n";

echo"item no:$row[0]  item name:$row[1]   unit price:$row[2]";}

}

}

?>
```

9. Write a PHP function to reverse a string

```
<html>

<head>

</head>

<body>

<form action="" method="POST">

enter string to reverse:<input type="text" name="rev"><br><br>

<input type="submit" value="submit">

</form>

</body>

</html>

<?php

if( $_POST )

{

$strrev = $_POST[ 'rev' ];

$strreverse = strrev($strrev);

echo "<b>reverse of a string is: </b>$strreverse<br>";

}

?>
```

10.Write a php script to upload the file to the server through HTML page.

```
<html>
 <body>
<form action = "" method = "POST" enctype = "multipart/form-data">
Select a File <input type ="file" name = "image" /><br>
<input type = "submit"/>
</form>
 </body>
</html>
<?php
if(isset($_FILES['image'])){
echo "File Name: ".$_FILES['image']['name']."<br>";
echo "File Size: ".$_FILES['image']['size']."<br>";
echo "File Type: ".$_FILES['image']['type']."<br>";
echo"<br>Success Uploaded.....";
}
?>
```

11. Create a table bank with the following fields accno, cust_name, balance. Insert some Records. Design HTML page to accept Account Number and amount using TextBoxes and type of transaction (Deposit or Withdrawal) using Radio Button. Write PHP program to update balance according to the transaction